

Eco-Loop Building Agents

Problem Statement ID – 1) TYPE - MS - Eco-Loop Building Agents

Problem Statement Title – Eco-Loop Building Agents

Theme – Building energy / Physical AI / Autonomous building agents

PS Category – Software

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Idea title: Eco-Loop Building Agents

One-line: Safe autonomous building optimisation through EnergyPlus, MCP and deterministic AI control

Links

- Live PoC: <https://twinpilot.webyaar.in>
- Code: <https://github.com/alphasafal/Twincity-/tree/ecoloop-hackathon-final>

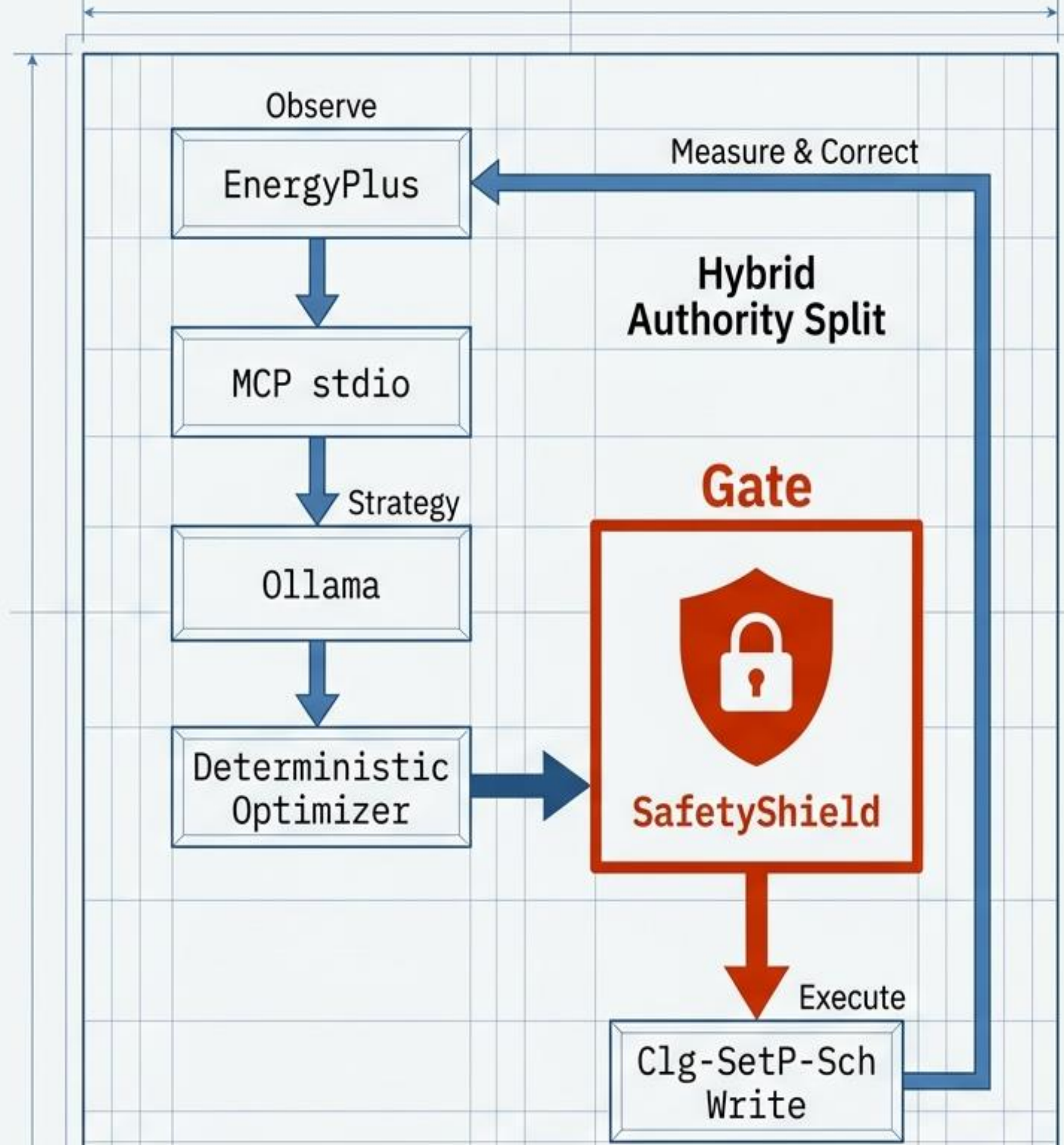
Proposed Solution

Problem Fit

- **Fixed schedules** → replaced by **state-aware control** (weather, occupancy, temps).
- **Energy waste** → measured HVAC 4.98% / total 1.31% savings.
- **Unsafe LLM control** → strictly gated by range, rate-limit, and fallback checks.

Innovation & Uniqueness

- **Hybrid authority split**: LLM strategizes, optimizer computes, SafetyShield dictates.
- **Real stdio MCP** (separate process PIDs) — no fake in-process tools.
- **Self-correction** via expected vs. actual outcome tracking.
- **Honest dual path**: Path A (savings proof) / Path C (LLM/MCP agency proof).



Technical Approach

Tech Stack Matrix

Simulation:	EnergyPlus 24.1 Runtime API (pyenergyplus)
Language:	Python 3.12 · TypeScript / Next.js 15
LLM:	Local Ollama (llama3.2:1b) – self-hosted
Tool Bus:	MCP over stdio (twinpilot-mcp)
Control:	Deterministic optimizer + SafetyShield
Models:	.idf office + Chicago .epw

Code Entry Points

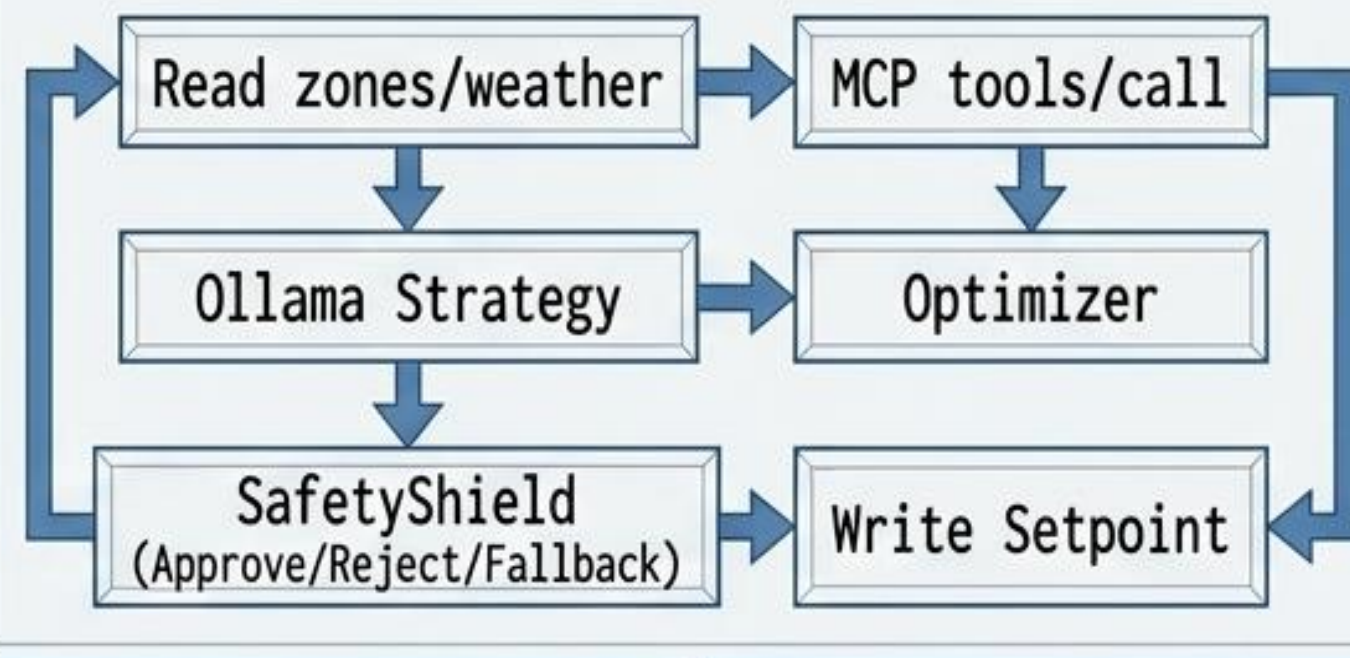
```
> simulator/ep_experiment.py (EnergyPlus loop)
> optimizer/safety.py (Safety parameters)
> twinpilot_mcp/energyplus_experiment_server.py
(MCP server)
> twinpilot_mcp/stdio_session.py (MCP client)
```

Methodology Loop

1. Baseline EnergyPlus run (fixed schedules)

2. Agent / hybrid run (identical parameters)

3. Hourly state loop:



4. Compare baseline vs. agent (% kWh + comfort)

5. Dashboard update (DATA_MODE=energyplus)

Feasibility and Viability

Feasibility

100% Open-source stack (no paid APIs), reproducible scripts, clean-clone verified.



Viability

Digital-building PoC validates logic prior to physical BMS integration.



Scalability

Scaffolded for BACnet / Modbus / Honeywell connectors.



Governance

Operators retain absolute approval authority (safe degraded modes).



Risk Mitigation Matrix

Identified Risk 	Deterministic Strategy 
LLM invents unsafe setpoints	SafetyShield hard gate; LLM never actuates.
LLM / Ollama server goes down	Deterministic fallback activates; simulation loop continues smoothly.
MCP fake / in-process claims	Real stdio implementation; separate process PIDs + protocol traces.
Comfort vs. energy savings tradeoff	Strict occupied caps; Path A claim strictly based on comfort-zero impact.
Over-claiming savings	Publish and measure verified Path A only.

Artifacts & Proof of Execution

Path A Verified Results (Comfort-Zero)			
Metric	Baseline	Eco-Loop	Δ (Delta)
HVAC	13.85 kWh	13.16 kWh	4.98%
Total	421.51 kWh	416.00 kWh	1.31%
Peak	19.93 kW	19.64 kW	1.47%
Comfort	0 h	0 h	-

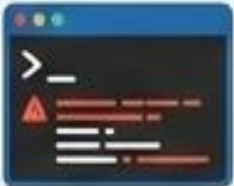
Artifact & Prototype Checklist

**Dashboard & Code:**
Live KPIs at twinpilot.webyaar.in/dashboard | Full GitHub monorepo branch.

**Trace & Protocol Evidence:**
Comparison JSON exports | MCP traces (client PID ≠ server PID) | Actuator CSV traces (Clg-SetP-Sch write = True).

**System Action Logs:**

- A. Safety rejection log (35°C blocked, no actuator write)
- B. Fallback activation log (Ollama down → deterministic_fallback x48)
- C. Hybrid Self-Correction (Expected vs. actual via self_correction.jsonl)



RESEARCH AND REFERENCES

Research & references (links)

EnergyPlus Runtime API — <https://energyplus.net>

Model Context Protocol (MCP) — <https://modelcontextprotocol.io>

Ollama (local OSS LLM) — <https://ollama.com>

Building energy context: buildings $\approx$ large share of global energy use (IEA / industry reports)

Project evidence pack — final-release/ + Eco-Loop-Hackathon-Submission.zip

Live PoC — <https://twinpilot.webyaar.in>

Source — <https://github.com/alphasafal/Twincity-/tree/ecoloop-hackathon-final>

AI proposes.

SafetyShield validates.

EnergyPlus executes.